



Figure 4: Where can vendors make money

Why you need a community

The first step towards building a community is a consortium. A consortium, defined simply, is a group made up of two or more individuals, companies, or governments that work together to achieve a common objective. The consortium is responsible for defining the format, as well as defining and publishing the reference architecture. A community adopts and owns such architectures, and makes them better.

The OPC Foundation defines how a machine will talk to another machine. The community then uses the framework given by OPC as a reference and comes out with standards like IEC62541. The community, in a nutshell, is responsible for coming out with frameworks that machines use to talk with each other.

The need and importance of a community can be better explained with

The community, in a nutshell, is responsible for coming out with frameworks that machines use to talk with each other.

the example of the English language. The framework that was defined hundreds of years ago is helping people from different countries and continents to interact with each other today. Similarly, the open source frameworks adopted by different machines and factories allow different components to generate and transmit data from one point to another.

You can go to any part of the world and converse with people if you know English as a language. Rulesets are needed, and communities play an essential role in letting different components adopt these rulesets. Then the community builds on top of these rulesets, gives feedback, and also makes sure that everybody follows that.

An open source community also helps in saving costs, as a company does not need to invest in developing new software. Knowing what exists in the market gives one the choice to explore and deploy the framework that best suits one's needs. Open source communities also allow building applications on top of existing frameworks. Companies can find out the way the software is written and APIs are defined, which can help in connecting all the components and making applications and usage better.

Independence, customisability and innovation are three more advantages of having an open source community. These three go hand-in-hand and allow companies the freedom to choose between components, system players, and vendors. This is where the power of open source comes in. Everything is open, including the customisations.

Brett Cannon, the Python core developer, said: "I came for the language (Python). I stayed for the community." This is a very powerful quote defining the value a community brings in. It goes without saying that the whole community is bigger than the sum of its parts. **END** 🐧

The article is based on a talk given by Bharath Sridhar, engineering manager, Industry 4.0, at Open Source India 2021.

By: Mukul Yudhveer Singh

The author is a business editor at Open Source For You.